Explainable Human Motion Computing with KSAS and XROCKET

Luca Foresti

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Abstract

This report applies XROCKET to the course-provided KSAS smartphone inertial dataset to study American Kenpo Karate Blocking Set I. The goal is not only to classify movement labels, but to answer where the information appears in the sensor axes, at what temporal scale it appears, and whether discriminative patterns can be interpreted from a human motion computing perspective. The final evaluation uses participant-grouped folds to avoid train-test participant leakage. The primary XROCKET random-forest model reached mean macro F1 0.883 over five folds; a logistic-regression sensitivity model reached mean macro F1 0.884. The strongest stable explanation evidence points to gravity and device-frame z-axis structure, long temporal spans in the padded 56-sample representation, and representative PPV pattern intervals that are plausible but not sufficient for validated coaching or biomechanical causation claims.

Introduction

Human Motion Computing treats movement as both an action and an information source. In this project, the action is a set of karate blocking movements, and the information source is a smartphone inertial measurement stream. The selected assignment scope is Part I - Human Motion Computing with Inertial Signals. The technical aim is to apply XROCKET, an explainable time-series classifier, to the KSAS dataset and provide evidence for three required questions:

1. Which sensor axes contribute most to movement classification?
2. Which temporal scales are most informative?
3. Which discriminative patterns appear meaningful from a human perspective?

XROCKET was selected because it exposes feature metadata such as channel, dilation, kernel length, and threshold (dida Data Science 2024a, 2024b). This metadata makes it possible to trace model behavior back to sensor-coordinate evidence instead of reporting only aggregate accuracy.

Dataset

The dataset is the course-provided KSAS smartphone IMU dataset (Human Motion Computing Course 2026). The audited local copy contains 240 CSV recordings: 20 participants, two arms, and six movement labels with 40 samples per label. The labels are no movement, upward block, hammering inward block, extended outward block, outward downward block, and rear elbow block. Each CSV contains 18 channels: accelerometer, gravity, gyroscope, linear acceleration, game rotation vector, and magnetic field, each with x, y, and z device-frame axes.

The audit found no missing values, no duplicate rows, no duplicate files, and no schema errors. Sequence lengths vary from 18 to 56 samples, with median length 33. Raw CSVs are kept under `data/raw/KSAS-Dataset/` locally but are ignored by Git and are not redistributed in the repository.

Methods

The repository builds a manifest from the audited raw files, converts the 240 recordings into a tensor with shape `(240, 18, 56)`, and right-pads shorter recordings with zeros. The tensor contract records sample order, channel order, labels, participant IDs, arm metadata, original lengths, and valid-timestep masks.

Evaluation uses five participant-grouped folds. All samples from a participant remain in the same fold, and each fold contains 48 held-out samples with eight samples per class. The reported metrics therefore test participant-independent generalization rather than sample-level interpolation.

Baselines use masked statistical features with majority, logistic-regression, and random-forest classifiers. The primary XROCKET experiment fits encoder thresholds on training folds only, extracts 9,072 PPV features per fold, and trains a random forest. A logistic-regression model on the same transformed features is retained as a sensitivity check.

The XROCKET implementation is pinned to `dida-do/xrocket` commit `1511e810c59d0c42f6431ef2f1f9fa57c71e9b2f` (dida Data Science 2024b). The public upstream repository had no public license file during this project, so this report treats its use as course-authorized academic support and does not claim a general open-source license for the upstream code.

Performance Summary

The statistical random-forest baseline reached mean macro F1 0.909, while the primary XROCKET random forest reached mean macro F1 0.883. This means the selected XROCKET representation is not the strongest classifier in the repository, but it is the main explainable model because it provides traceable channel and dilation metadata.

Model	Mean macro F1	Mean balanced accuracy	Notes
Majority baseline	0.048	0.167	Sanity baseline
Statistical logistic regression	0.886	0.888	Masked statistical features
Statistical random forest	0.909	0.908	Strongest baseline
XROCKET random forest	0.883	0.883	Primary explainable model
XROCKET logistic regression	0.884	0.883	Sensitivity model

The primary XROCKET random forest was weakest for labels 2 and 3, with recalls 0.750 and 0.775. These are the hammering inward block and extended outward block, and their mutual confusion is discussed as a limitation.

Task 1.1 - Sensor-Axis Contribution

Answer to Task 1.1

The strongest stable sensor-axis evidence comes from the gravity sensor family and the pooled device-frame z axis. Native feature importance ranked gravity first among sensor families with mean normalized importance 0.200, followed by gyroscope at 0.175 and accelerometer at 0.168. The pooled z axis ranked first with mean normalized importance 0.347. At the individual-channel level, `gravity_z`, `gravity_x`, `gravity_y`, `gyros_x`, `game_rot_vec_z`, and `accelerometer_z` were among the leading channels, but channel-level claims are weaker because M7 showed top-channel instability across seed/fold cases.

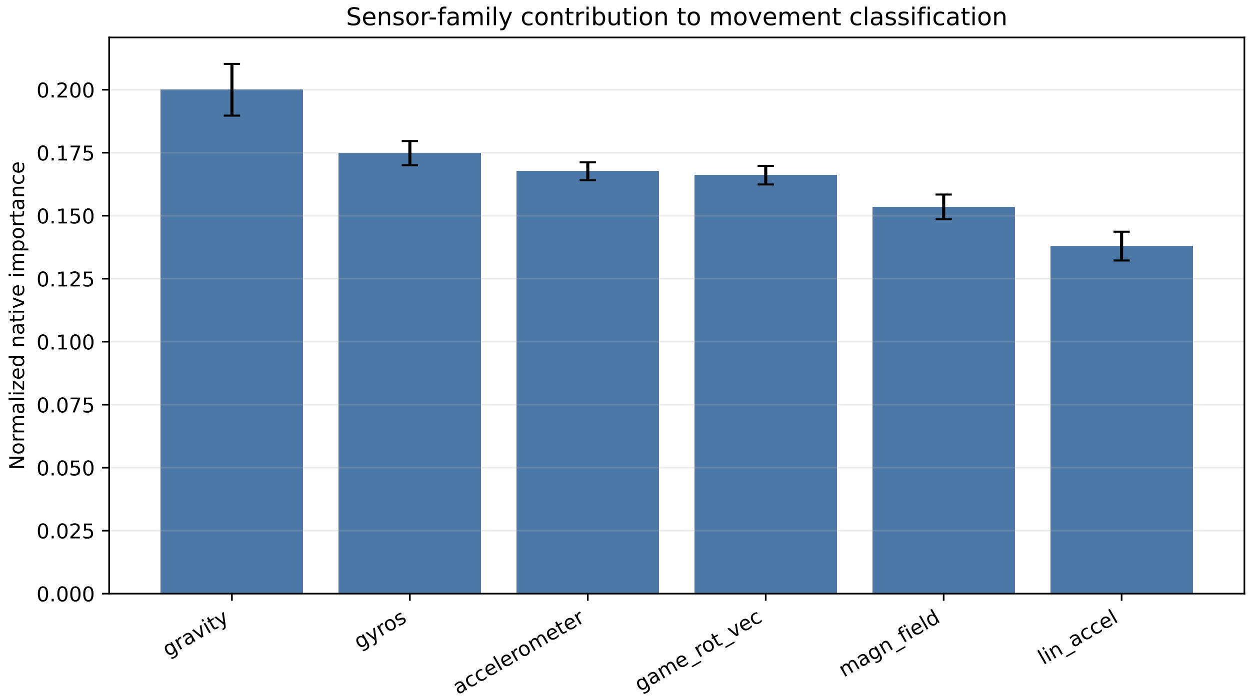


Figure 1: Task 1.1 sensor-family contribution. Gravity has the highest native importance and is also supported by grouped permutation evidence.

Validation evidence is mixed but useful. Grouped test-set permutation produced the largest sensor-family macro-F1 drop for gravity, followed by game rotation vector and gyroscope. Feature-group ablation produced small or negative average drops, which indicates redundancy in the transformed feature bank rather than a simple one-sensor dependency.

Biomechanically, gravity and rotation-related evidence is plausible for blocking motions because the phone is worn on the forearm and the movements change forearm orientation and acceleration. However, all axes are Android device-frame axes. Without anatomical calibration, left/right mirroring checks, or expert-labeled movement phases, this report avoids claims about exact joint mechanics, muscle activation, force, or skill quality.

Task 1.2 - Temporal-Scale Analysis

Answer to Task 1.2

The saved padded XROCKET representation relies primarily on long-duration patterns. With kernel length 9, the available dilations map to receptive-field spans of 9, 17, 25, 33, 41, and 49 samples. Long spans, defined as 41 and 49 samples in the 56-sample padded window, contributed mean normalized importance 0.655. Intermediate spans contributed 0.224, and short spans contributed 0.122. The leading dilation was 6, corresponding to a 49-sample receptive field.

Approximate seconds are reported only as a nominal conversion. The KSAS app requested Android `SENSOR_DELAY_GAME`, treated here as approximately 50 Hz, but the CSV exports do not retain event timestamps. Android sensor delays are requests rather than guaranteed exact acquisition intervals (Android Developers 2026). Therefore, the primary temporal evidence is in samples: 41 and 49 samples, not verified physical seconds.

The result suggests that recognition depends more on broad movement structure than on very local impulses.

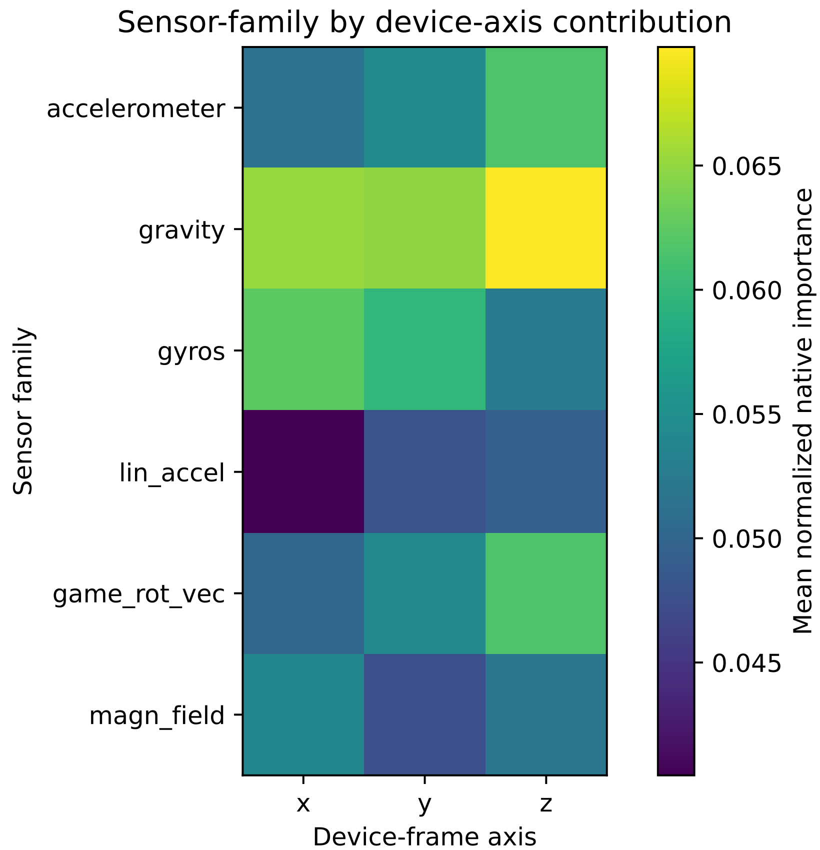


Figure 2: Task 1.1 device-frame channel and axis heatmap. Values are normalized within folds and should be read as relative model-use evidence.

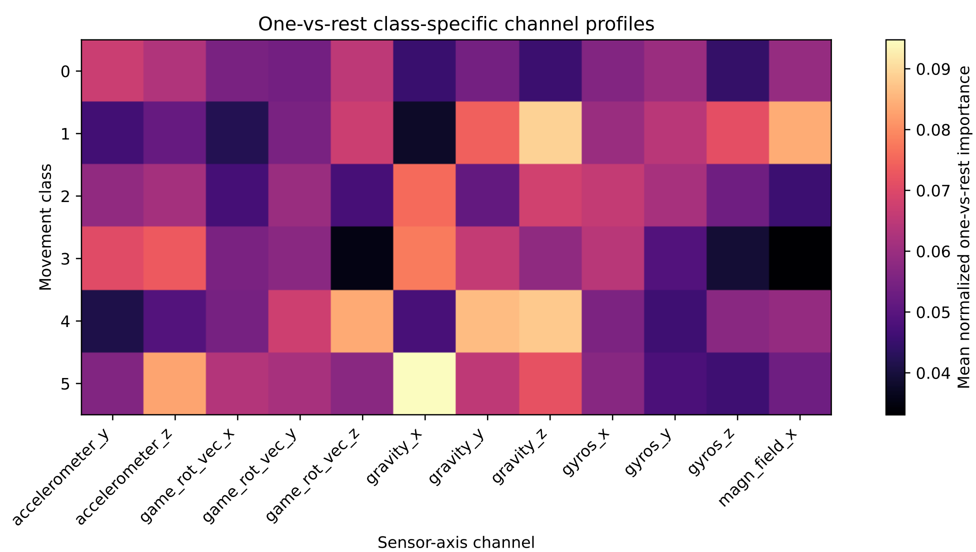


Figure 3: Task 1.1 class-specific channel profiles. These one-vs-rest diagnostics show movement-specific channel emphasis but are secondary evidence.

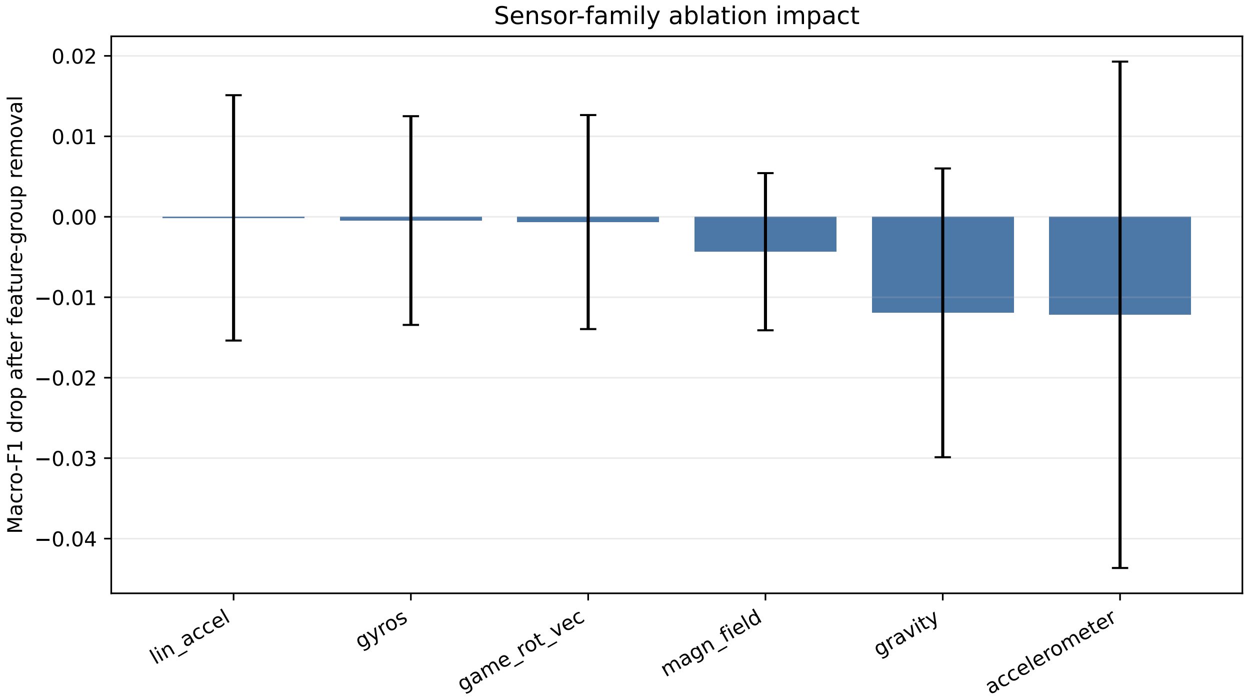


Figure 4: Task 1.1 ablation impact. Small and sometimes negative drops indicate redundancy among transformed features.

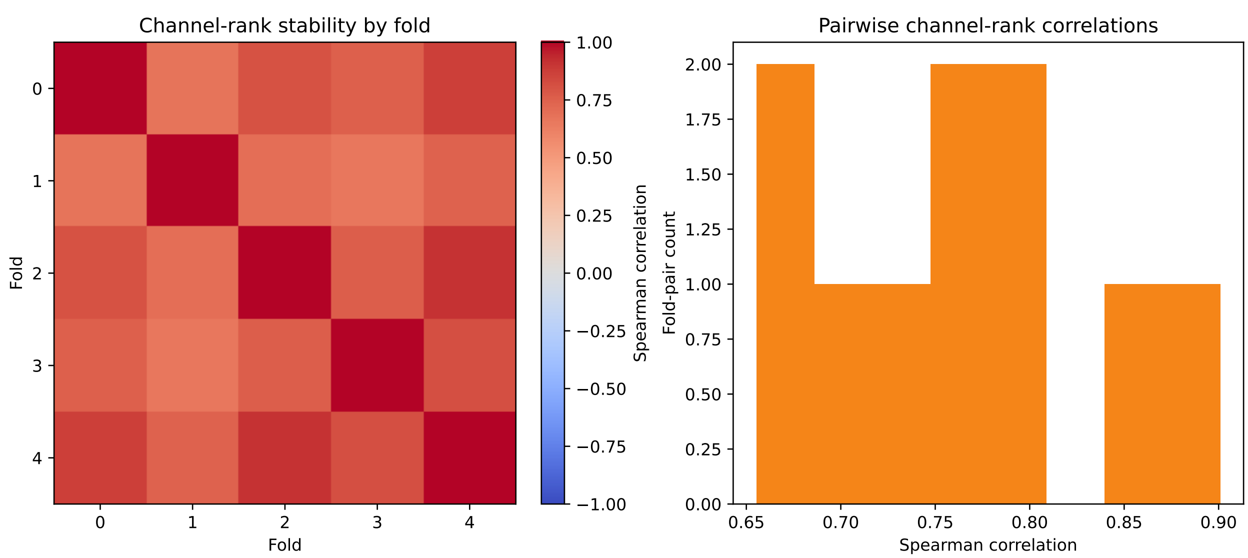


Figure 5: Task 1.1 fold stability. Stable family-level findings are stronger than individual-channel claims.

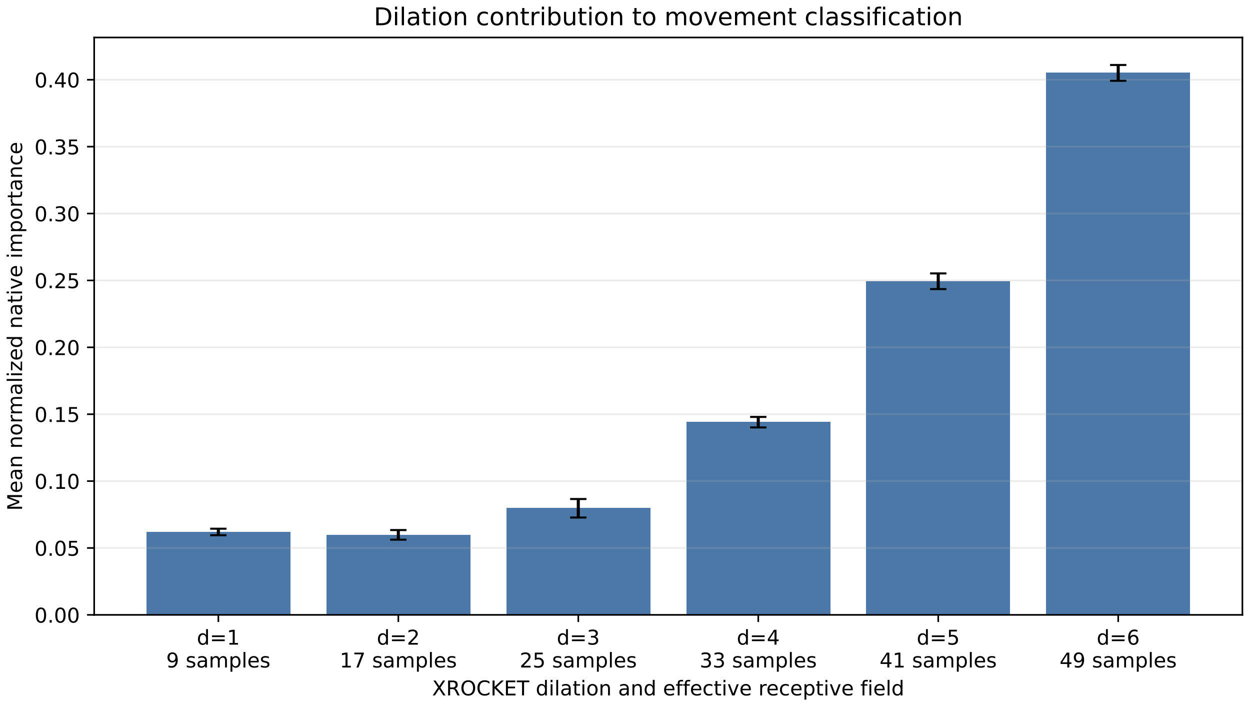


Figure 6: Task 1.2 dilation importance. Dilation 6 is the strongest contributor in the saved padded XROCKET representation.

This is plausible for karate blocking gestures, where the full arm trajectory and orientation transition may distinguish classes. The caveat is important: XROCKET has no valid-timestep mask, so the temporal explanation describes the padded representation. Padding diagnostics showed prediction and feature sensitivity, so a mask-aware or timestamped representation would be needed before making stronger time-domain claims.

Task 1.3 - Discriminative-Pattern Interpretation

Answer to Task 1.3

The most discriminative XROCKET patterns can be mapped to representative signal intervals, but the localization is approximate. The selected features are PPV features: each transformed value is the proportion of convolution responses above a learned threshold. The interval shown in each case is therefore the strongest representative above-threshold segment, not a unique causal instant.

The selected stable features were mostly long-span patterns at dilation 5 or 6. The first cases involved `lin_accel_y`, `gravity_z`, and `lin_accel_z`, with associated movement labels upward block and hammering inward block. Two cases were labeled plausible from a human perspective, and two were labeled ambiguous.

The patterns appear meaningful as sensor-coordinate evidence of broad movement shape, orientation change, and acceleration structure. They could support a future human-in-the-loop review workflow by highlighting signal regions used by the model. They do not validate a coaching system, performance scoring, expertise assessment, force estimate, or learning-gain claim.

Robustness, Confounds, And Problems Encountered

M7 controls found no unresolved evidence of participant leakage or obvious label leakage. Label-shuffle controls stayed low: the strongest mean macro F1 was 0.1605 and the maximum fold/seed macro F1 was

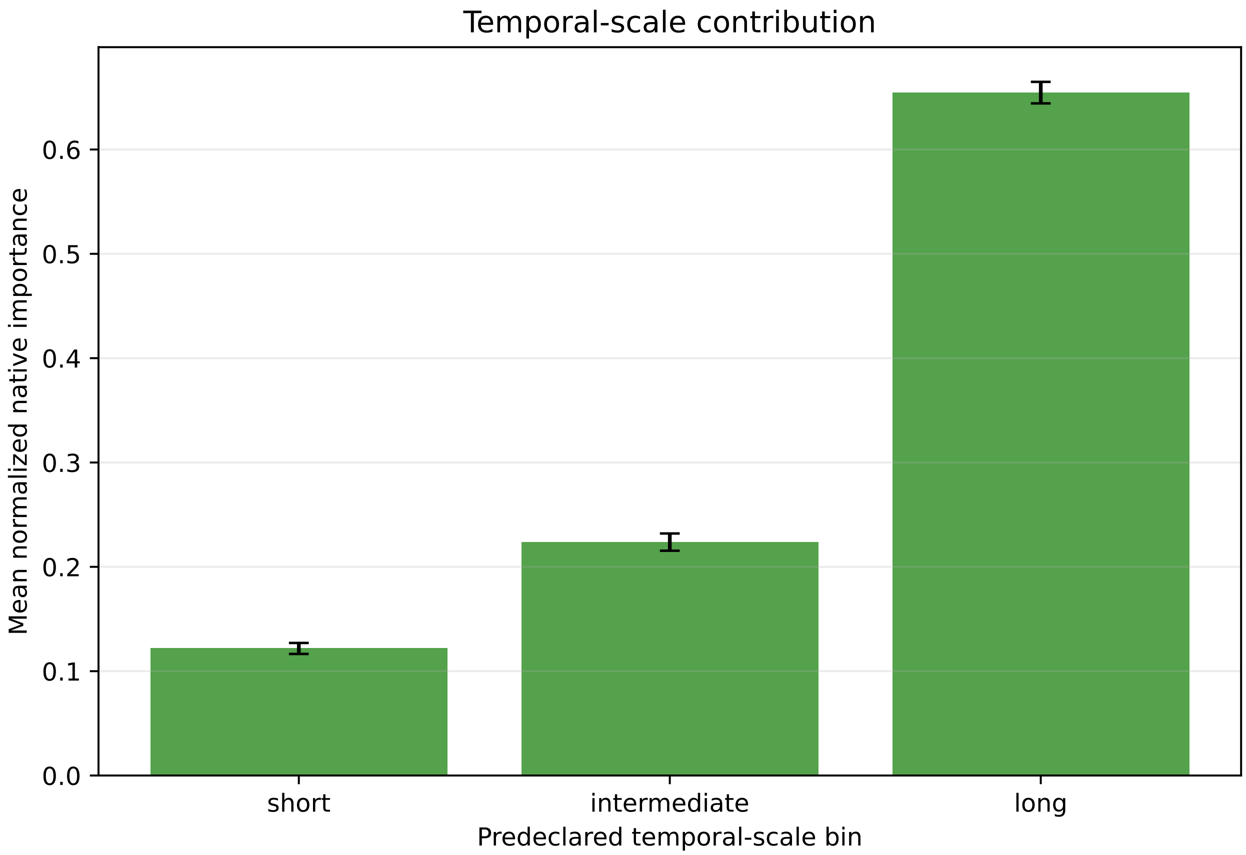


Figure 7: Task 1.2 temporal-scale contribution. Long spans dominate the normalized importance profile.

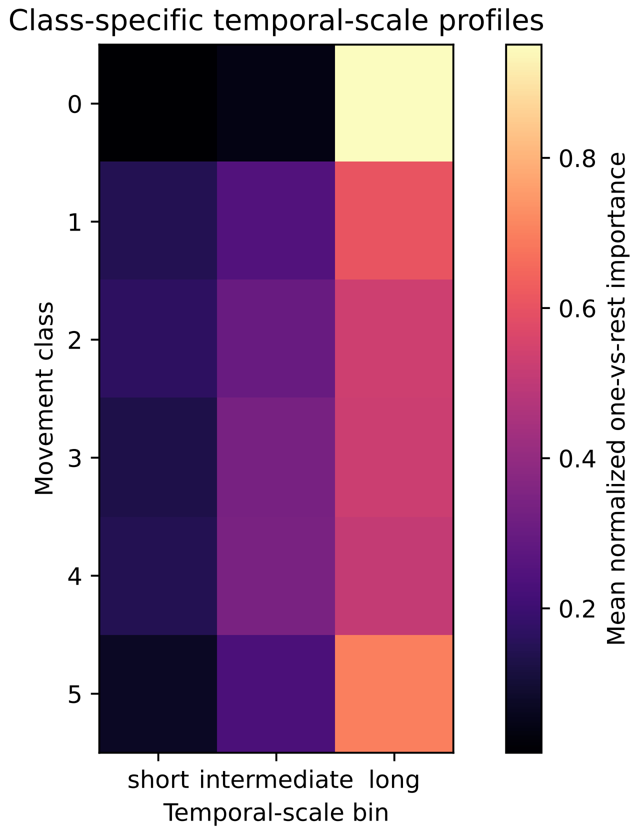


Figure 8: Task 1.2 class-specific temporal-scale profiles. Most classes show long-scale dominance, with class 4 closest to mixed evidence.

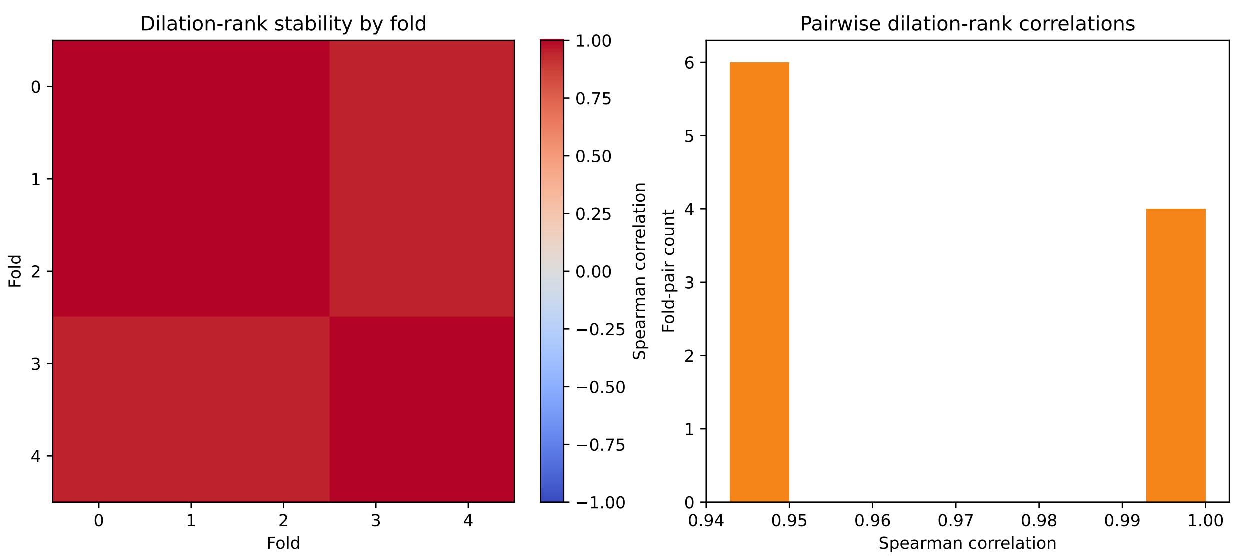
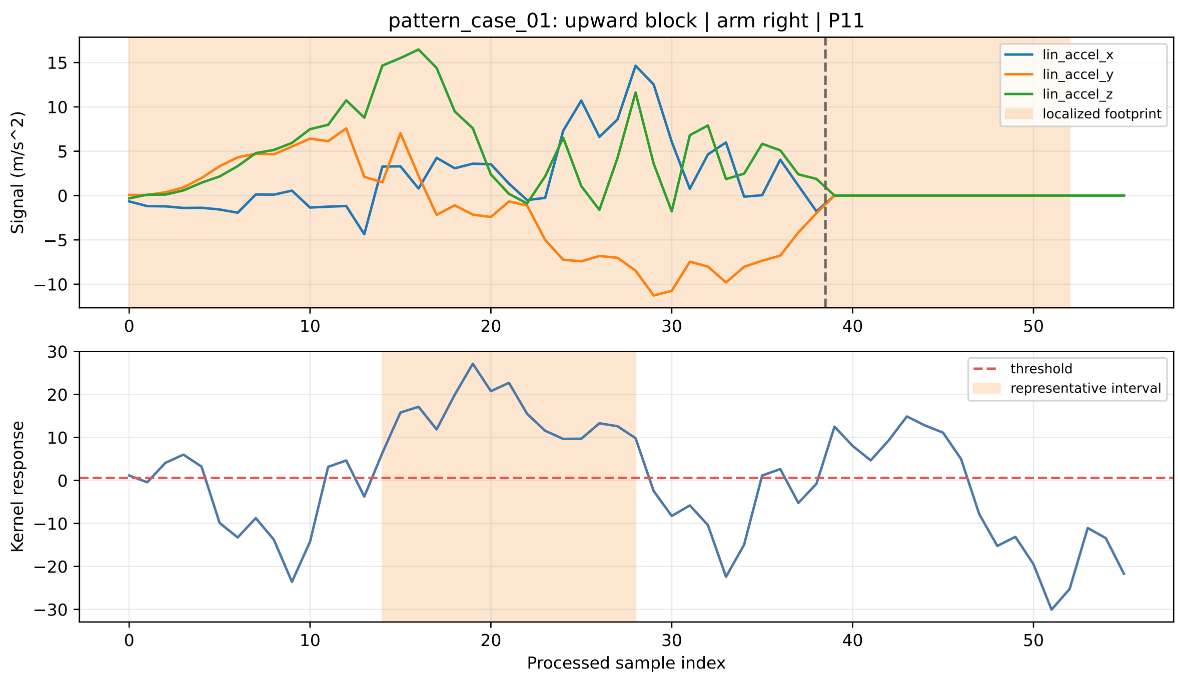
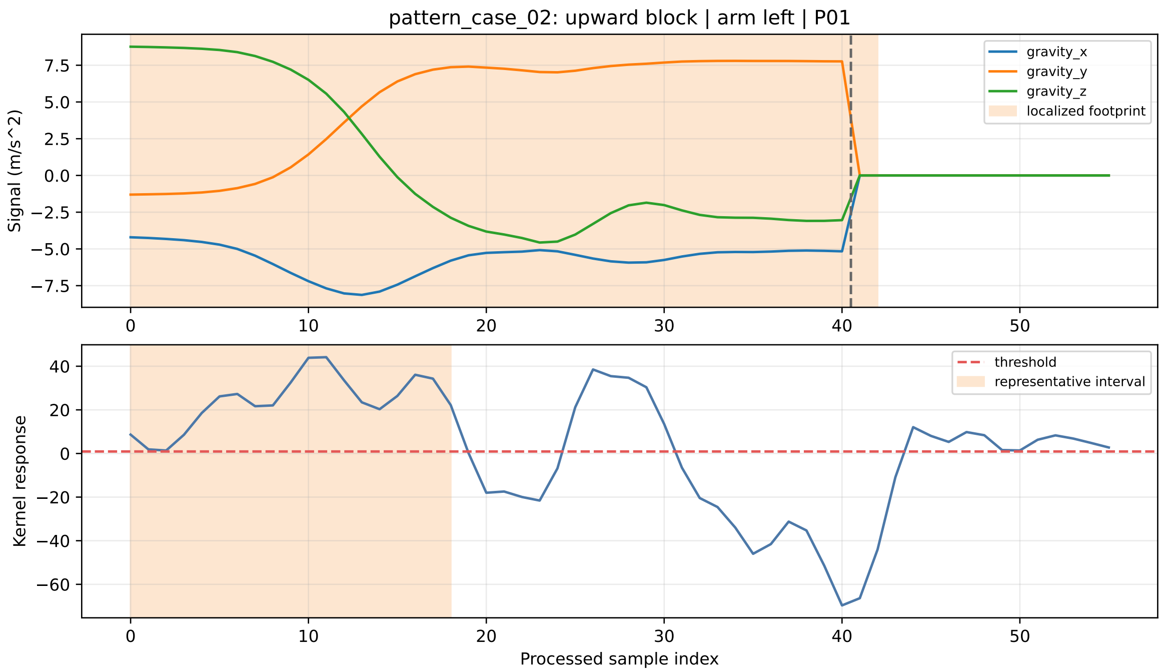


Figure 9: Task 1.2 fold stability. The long-scale conclusion is stable at the main scale-bin level.



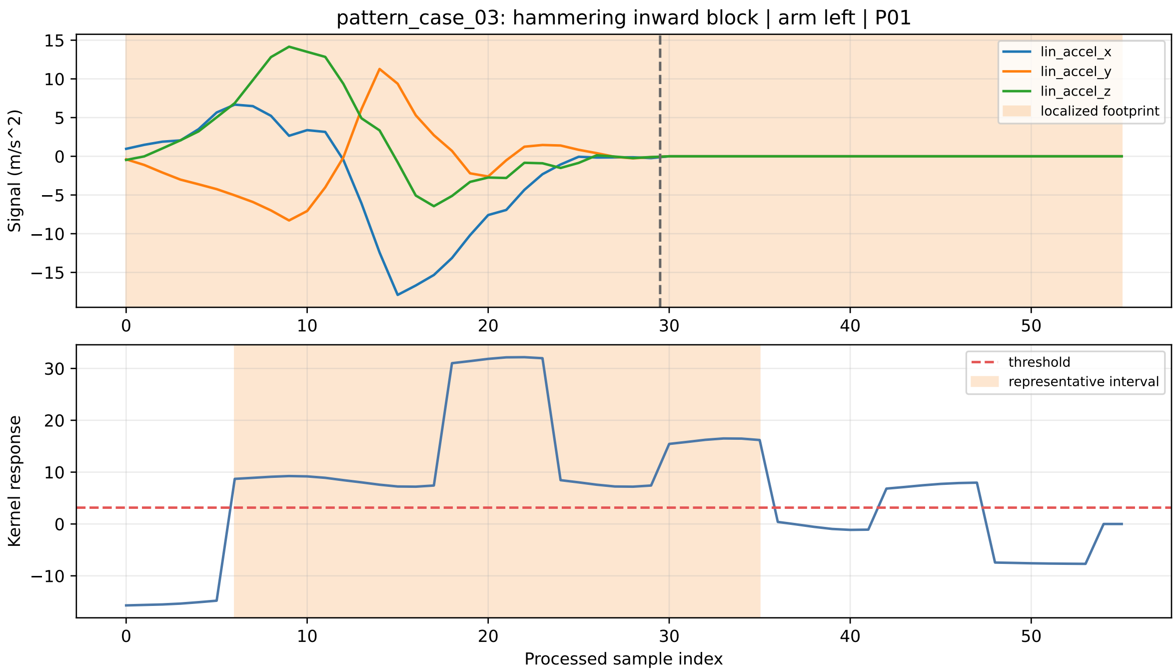
Feature 8164 | d05_p033 | dilation 6 | span 49 samples
PPV=0.554, AUC=0.682, meaningfulness=plausible
True=1 Pred=1 Margin=0.560
This long-span PPV pattern captures gravity-removed linear acceleration on the device y-axis over about 49 samples. It may reflect broad movement shape, transition timing, or sustained orientation/velocity structure, but only in sensor coordinates. The interval overlaps padded samples, so the meaning is representation-dependent.

Figure 10: Task 1.3 pattern case 1. Representative correct case for a high-importance PPV feature.



Feature 8429 | d05_p048 | dilation 6 | span 49 samples
PPV=0.661, AUC=0.971, meaningfulness=plausible
True=1 Pred=1 Margin=0.682
This long-span PPV pattern captures orientation-related gravity structure on the device z-axis over about 49 samples. It may reflect broad movement shape, transition timing, or sustained orientation/velocity structure, but only in sensor coordinates. The interval overlaps padded samples, so the meaning is representation-dependent.

Figure 11: Task 1.3 pattern case 2. A second correctly classified representative pattern case.



Feature 7931 | d05_p020 | dilation 6 | span 49 samples
PPV=0.643, AUC=0.822, meaningfulness=ambiguous
True=2 Pred=2 Margin=0.446
This long-span PPV pattern captures gravity-removed linear acceleration on the device z-axis over about 49 samples. It may reflect broad movement shape, transition timing, or sustained orientation/velocity structure, but only in sensor coordinates. The interval overlaps padded samples, so the meaning is representation-dependent.

Figure 12: Task 1.3 pattern case 3. A correctly classified case with more ambiguous human meaning.

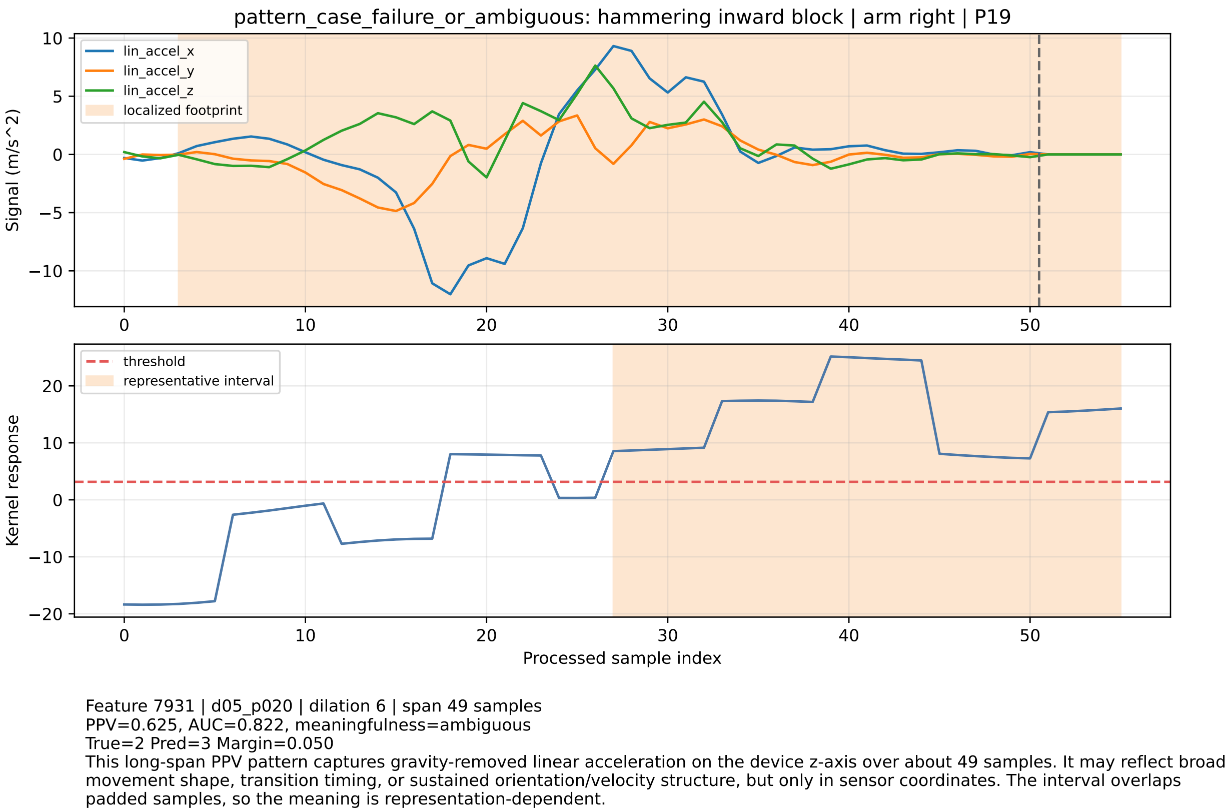


Figure 13: Task 1.3 failure or ambiguous case. Retaining this case makes the explanation limits visible.

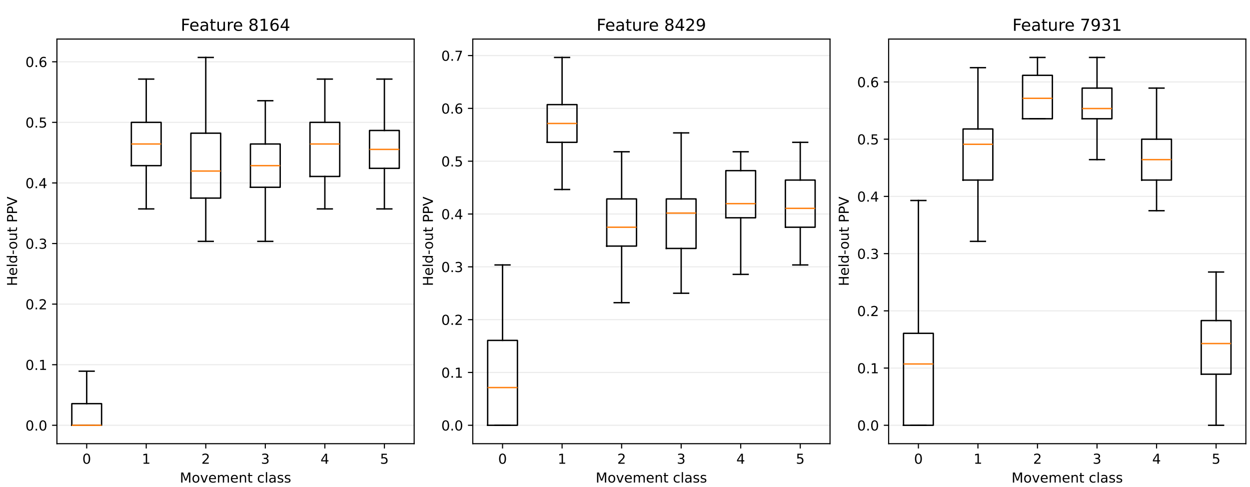


Figure 14: Task 1.3 feature distributions. Class separation supports the selected patterns but does not prove causation.

Task 1.3 pattern summary

case	movement	channel	dil.	PPV	AUC	meaning
case 01	upward block	lin_accel_y	6	0.554	0.682	plausible
case 02	upward block	gravity_z	6	0.661	0.971	plausible
case 03	hammering inward block	lin_accel_z	6	0.643	0.822	ambiguous
ambiguous	hammering inward block	lin_accel_z	6	0.625	0.822	ambiguous

Figure 15: Task 1.3 pattern summary table. Human-meaningfulness labels are cautious report aids, not expert biomechanical validation.

0.3151. Metadata-only controls using sequence length, padding fraction, and arm code also stayed low, with the strongest mean macro F1 0.2158.

The main problems encountered were:

- raw CSVs had no timestamps, so realized sampling rate and jitter are unknown;
- sequence lengths varied, requiring right-padding to 56 samples;
- XROCKET has no valid-timestep mask, so padding affects some predictions and temporal explanations;
- feature metadata required a local adapter and tests because flattened feature order had to be aligned with saved feature columns;
- channel-level rankings were less stable than family, axis, and temporal-scale rankings;
- labels 2 and 3 were the weakest classes and are mutually confused;
- Quarto was not installed locally, so the final report uses Pandoc with MiKTeX/pdflatex instead of a Quarto pipeline.

Limitations And Threats To Validity

The dataset is small: 240 samples from 20 participants. Results are descriptive and assignment-focused, not a clinical, coaching, diagnostic, or injury-prevention validation. Participant-independent folds reduce leakage, but they do not prove generalization beyond the KSAS protocol.

Axis interpretation is limited by phone placement and device coordinates. The phone was intended to be worn along the forearm, but participant-specific placement and orientation consistency were not available. Arm-stratified checks found that the top pooled axis differs by arm code, so pooled z-axis evidence must be read with an arm-orientation caveat.

Temporal interpretation is limited by missing timestamps and padding. Receptive fields are reported in samples first, with approximate nominal-50-Hz seconds only as context. Pattern interpretations are heuristic and must not be treated as validated biomechanics.

Reproducibility And Repository

Repository URL: https://github.com/Forest904/ksas-xrocket-hmc.git

Submission identifier: Git tag `v1.0-submission` on branch `main`.

The canonical Python environment is `pyproject.toml` plus `uv.lock` targeting Python 3.11. Raw KSAS data must be placed locally under `data/raw/KSAS-Dataset/`; raw participant CSVs are ignored by Git.

The compact final checks are:

```
uv sync --locked
make reproduce
make figures
make report
pdftoppm -png reports/ksas_xrocket_hmc_report.pdf tmp/pdfs/ksas_report
```

The full artifact rebuild sequence is documented in `docs/reproducibility.md`. Some commands require `--overwrite` if rerunning over existing result directories. The committed `results/` directory contains the final artifacts used by this report.

No new repository license is added in this submission. Original project code is provided for academic assignment review, raw KSAS data are excluded, and XROCKET is attributed as pinned course-authorized upstream software without a public upstream license file.

Generative-AI Disclosure

This Generative-AI disclosure covers retained repository, software, and report work. Codex was used between 9 June and 9 July 2026 to support repository planning, software implementation, debugging, test design, documentation updates, report planning, and language editing. Generated code and prose retained in the project were reviewed, executed, tested, and revised by the author. Raw participant data, secrets, and personally identifying information were not supplied to generative-AI tools. The author retained responsibility for final decisions, interpretation, verification, and submission.

Conclusion

This project produced a reproducible XROCKET analysis and final PDF report for the KSAS inertial dataset. The main scientific conclusion is cautious: the model uses gravity and device-frame orientation/acceleration structure, relies most strongly on long spans in the padded XROCKET representation, and exposes representative pattern intervals that can guide expert review. The evidence is useful for explainable human motion computing, but it remains sensor-coordinate and protocol-specific rather than validated biomechanical or pedagogical truth.

References

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